

YouTube Trending Videos Analysis

Data Visualization Project Report | Emy Kirugo

This report presents three data visualizations built in Tableau Public, exploring the US YouTube Trending Videos dataset. Each video's cumulative statistics were de-duplicated to reflect only its final (last) day of trending, ensuring likes, dislikes, and view counts are not overstated across a video's multi-day trending run.

Insight 1: Trending Videos by State

Link: https://public.tableau.com/shared/J97XWJQFH?:display_count=n&:origin=viz_share_link

Summary of Insight

This map shows how many distinct trending videos originated from each US state between November 2017 and March 2018. California leads by a wide margin with 567 trending videos, followed by Texas (433), Georgia (369), Illinois (330), and Florida (321). While the top states generally track with population size, Georgia stands out: it ranks 8th in US population but 3rd in trending videos, ahead of larger states like New York. This lines up with Atlanta's well-documented role as a major hub for the hip-hop, comedy, and film/TV industries. For anyone studying where online content and creator activity concentrates, this map suggests Atlanta functions as a media capital on par with more traditionally recognized hubs like Los Angeles and New York.

Design Choice Description

A filled (choropleth) map was chosen because state-level geographic comparison is the clearest way to answer a location-based question, letting the reader instantly see regional concentration without needing to read a table of 50 rows. The color palette is a single-hue sequential scale (light to dark orange) rather than a diverging red/green scale, since the underlying measure is simply a count with no inherent 'good' or 'bad' direction, since a diverging palette would incorrectly imply a value judgment. The tooltip was customized to clearly label Country, State, and video count so the exact figure behind each state's shading is always available on hover.

Insight 2: Category Likes vs. Dislikes

Link:https://public.tableau.com/views/YouTubeTrendingVideosAnalysis_17853905656380/CategoryLikesvsDislikes?:language=en-US&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link

Summary of Insight

This chart compares total likes and dislikes across all 16 video categories present in the trending data. Music dominates likes with roughly 68.7 million, more than double the runner-up, Entertainment, at about 34.0 million. Comedy follows in third with about 19.3 million likes. The dislike totals tell a more interesting story: despite having far fewer likes than Music, Entertainment accrued more total dislikes (about 3.19 million versus Music's 2.27 million). In other words, Entertainment content draws a noticeably more polarized audience reaction relative to its popularity than Music does. For content creators or brands deciding where to invest, this suggests Music is both the highest-reach and the most consistently well-received category, while Entertainment content (which often includes late-night TV, celebrity news, and opinion-driven segments) should be expected to generate more friction alongside its reach.

Design Choice Description

A horizontal bar chart split into two independently scaled panes (one for Likes, one for Dislikes) was used instead of a single shared axis, because Likes are measured in the millions while Dislikes are measured in the thousands, so a single shared scale would have rendered the Dislikes bars invisible. Each measure was assigned a single solid, distinct color (teal for Likes, pink for Dislikes) rather than a red/green scheme, since more dislikes is not necessarily a negative outcome in this context (it can simply reflect a more actively engaged or opinionated audience), so a traffic-light color scheme was intentionally avoided. Categories are sorted in descending order of Likes so the most-liked category is immediately visible at the top of the chart. An interactive Category Name filter was added so a reader can isolate one or more specific categories at a time, rather than only viewing the full 16-category comparison.

Insight 3: Real-World Events Drive Tag Spikes (Dashboard)

Link:https://public.tableau.com/views/YouTubeTrendingVideosAnalysis_17853905656380/TagTrendsDashboard

Summary of Insight

This line chart tracks the weekly popularity of six selected tags across the full trending window (mid-November 2017 to mid-March 2018). Three of the tags, 'Grammys', 'super bowl', and 'Olympics', are tied to specific real-world events, while the remaining three ('NFL', 'music', 'science') serve as an evergreen baseline for comparison. The results show a clear pattern: each event-tied tag spikes sharply in the exact week of its real-world event, then falls off almost as quickly. The 'Grammys' tag rises through late January and peaks right at the 2018 Grammy Awards (January 28). The 'super bowl' tag spikes even more sharply the following week, aligned with Super Bowl LII (February 4). The 'Olympics' tag builds steadily and stays elevated through the full 2018 Winter Olympics window (February 9-25) before dropping off. By contrast, the

evergreen tags 'music' and 'science' rise and fall on their own independent schedule, with no single triggering moment. This is a useful reminder for anyone using YouTube trending data to predict content demand: interest in a topic tied to a live event is extremely short-lived and concentrated, while interest in an evergreen topic is more diffuse but also more durable.

Design Choice Description

A multi-line time series was the natural choice for showing how interest in several topics' changes week over week. Rather than relying on an automatic 'Top N tags' ranking, which would have surfaced six near-duplicate synonyms for comedy content ('funny', 'comedy', 'humor', 'jokes', 'funny video', 'joke'), a manually curated set of six tags was used instead, chosen specifically to contrast event-driven spikes against evergreen baselines. Each tag was assigned its own color via a categorical palette, so the six lines remain visually distinct. Three labeled vertical reference lines were added at the exact dates of the Grammy Awards, Super Bowl LII, and the Olympics opening ceremony, turning the chart from a purely descriptive 'here are some lines' visual into an explanatory one that directly shows cause and effect between calendar events and platform behavior. The line's default color palette was replaced with Tableau's built-in Color Blind palette, so the six tags remain distinguishable for readers with color vision deficiency. This chart was combined into a Dashboard alongside a small companion table listing the exact date of each real-world event, so a reader can immediately cross-reference a spike on the chart with the specific event that caused it, without needing to guess from the reference line position alone. An interactive tag filter (inherited from the underlying Featured Tags set) is also exposed on the dashboard, letting a reader toggle individual tag on or off.

Resources

Dataset: YouTube Trending Video Statistics (Kaggle, [datasnaek/youtube-new](#)), as provided through the Udacity project materials, including the supplementary tag-transposition and category-name lookup files provided for this project. No other external websites, books, forums, or repositories were referenced for this submission.

N/A, beyond the dataset and course materials cited above.